

CBSE Class 12 Pe — Half-Yearly Predicted Paper (Set 5 of 10)

SECTION A: Objective Type Questions (18 Marks: 18 x 1)

All questions are compulsory. Consists of MCQs and Assertion-Reason questions.

Q1. Which tournament format is popularly known as a 'Round Robin' tournament? (A) Knock-out Tournament (B) League Tournament (C) Challenge Tournament (D) Consolation Tournament *Unit 1: Management of Sporting Events | Tag: Very High***

Q2. The mathematical formula to calculate the exact number of matches in a Knock-out tournament is: (A) N (B) $N - 1$ (C) $N + 1$ (D) $\frac{N(N-1)}{2}$ *Unit 1: Management of Sporting Events | Tag: Very High***

Q3. The spinal postural deformity characterised by a lateral (sideways) "S" or "C" shaped curve of the spine is termed as: (A) Lordosis (B) Kyphosis (C) Scoliosis (D) Flat foot *Unit 2: Children & Women in Sports | Tag: Very High***

Q4. A severe medical condition in female athletes characterized by the complete absence or cessation of menstrual periods for three or more consecutive months is called: (A) Menarche (B) Amenorrhea (C) Dysmenorrhea (D) Osteoporosis *Unit 2: Children & Women in Sports | Tag: Very High***

Q5. Which lifestyle disease is primarily characterized by the chronic inflammation, swelling, and narrowing of the respiratory airways, producing extra mucus? (A) Diabetes (B) Hypertension (C) Asthma (D) Obesity *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | Tag: Very High***

Q6. The official oath: "Let me win. But if I cannot win, let me be brave in the attempt" is recited by the athletes of which international organization? (A) Paralympics (B) Special Olympics (C) Deaflympics (D) Commonwealth Games *Unit 4: Physical Education & Sports for CWSN | Tag: High***

Q7. The expanded form of SPD, a condition where the brain has trouble receiving and responding to information that comes in through the senses, is: (A) Sensory Processing Disorder (B) Sensory Personal Disorder (C) Sensory Physical Disability (D) Sensory Problem Disorder *Unit 4: Physical Education & Sports for CWSN | Tag: Very High***

Q8. Which of the following vitamins belongs to the 'Water-Soluble' category and cannot be stored in the body for long periods? (A) Vitamin A (B) Vitamin D (C) Vitamin B-Complex (D) Vitamin K *Unit 5: Sports & Nutrition | Tag: Very High***

Q9. A severe deficiency of which essential mineral in the diet restricts the formation of hemoglobin, leading to Anemia? (A) Calcium (B) Iodine (C) Iron (D) Phosphorus *Unit 5: Sports & Nutrition | Tag: Very High***

Q10. In the SAI Khelo India Fitness Test battery for girls (9-18 years), which specific test is conducted to measure upper body strength and endurance? (A) Pull-ups (B) Push-ups (C) Modified Push-ups (D) Partial Curl-up *Unit 6: Test and Measurement in Sports | Tag: Very High***

Q11. The 'Chair Stand Test' in the Rikli & Jones Senior Citizen Fitness battery primarily evaluates: (A) Upper body flexibility (B) Lower body strength (C) Agility and dynamic balance (D) Aerobic endurance *Unit 6: Test and Measurement in Sports | Tag: Very High***

Q12. The total volume of blood pumped by the left ventricle of the heart in one full minute is scientifically known as: (A) Stroke Volume (B) Cardiac Output (C) Tidal Volume (D) Vital Capacity *Unit 7: Physiology & Injuries in Sport | Tag: Very High***

Q13. A soft tissue injury involving the severe stretching or tearing of a muscle or its tendon is medically classified as a: (A) Sprain (B) Strain (C) Contusion (D) Laceration *Unit 7: Physiology & Injuries in Sport | Tag: Very High***

Q14. The anatomical lever system where the Fulcrum (axis) is located strictly between the Effort (force) and the Load (resistance) is called a: (A) Class I Lever (B) Class II Lever (C) Class III Lever (D) None of the above *Unit 8: Biomechanics and Sports | Tag: Very High***

Q15. The resistance force that opposes the motion of a rolling object (like a football rolling on the grass) is known as: (A) Static friction (B) Sliding friction (C) Rolling friction (D) Fluid friction *Unit 8: Biomechanics and Sports | Tag: High***

Q16. Assertion (A): Consolation tournaments are specifically organized to provide a second opportunity to teams that get eliminated in the early rounds. **Reason (R):** A consolation tournament is a specific variation of the League tournament format. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 1: Management of Sporting Events | Tag: High***

Q17. Assertion (A): Roughage (dietary fiber) is considered a highly important non-nutritive component of a balanced diet. **Reason (R):** Roughage provides zero calories or energy to the body but is crucial for preventing constipation and ensuring smooth bowel movements. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 5: Sports & Nutrition | Tag: Very High***

Q18. Assertion (A): A shot-putter uses maximum explosive force to throw the iron ball into the air, but it eventually traces a parabolic path and falls to the ground. **Reason (R):** Gravity is the invisible downward force that continuously acts on all projectiles in flight, pulling them back toward the earth's surface. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 8: Biomechanics and Sports | Tag: Very High***

SECTION B: Very Short Answer Type (10 Marks: Attempt any 5 out of 6 x 2)

Word limit: 60-90 words.

- Q19.** Outline any two distinct objectives of organizing 'Extramural' sports tournaments. *Unit 1: Management of Sporting Events | Tag: Very High***
- Q20.** State any two corrective physical activities or exercises for the postural deformity known as 'Flat Foot'. *Unit 2: Children & Women in Sports | Tag: High***
- Q21.** Mention any two contraindications (who should avoid it) of practicing *Chakrasana* (Wheel Pose). *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | Tag: High***
- Q22.** Differentiate clearly between 'Classification' and 'Divisioning' in the context of adaptive sports. *Unit 4: Physical Education & Sports for CWSN | Tag: High***
- Q23.** Briefly explain the administration procedure of the 'Sit and Reach Test' used to measure an athlete's flexibility. *Unit 6: Test and Measurement in Sports | Tag: Very High***
- Q24.** Differentiate strictly between 'Vital Capacity' and 'Residual Volume' in the human respiratory system. *Unit 7: Physiology & Injuries in Sport | Tag: Very High***
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SECTION C: Short Answer Type (15 Marks: Attempt any 5 out of 6 x 3)

Word limit: 100-150 words.

- Q25.** Describe the specific roles and responsibilities of the 'First Aid Committee' and the 'Finance Committee' during the organization of a sports tournament. *Unit 1: Management of Sporting Events | Tag: Very High***
- Q26.** Discuss any three psychological benefits that women gain by actively participating in competitive sports. *Unit 2: Children & Women in Sports | Tag: High***
- Q27.** What are 'Food Myths'? State any two prevalent food myths regarding nutrition and weight loss, and explain their actual scientific facts. *Unit 5: Sports & Nutrition | Tag: Very High***
- Q28.** Detail the purpose, equipment required, and administration procedure of the 'Plate Tapping Test' designed for children aged 5-8 years. *Unit 6: Test and Measurement in Sports | Tag: Very High***
- Q29.** Elaborate on any three long-term physiological effects of regular endurance exercise on the muscular system (e.g., muscle hypertrophy, mitochondrial density). *Unit 7: Physiology & Injuries in Sport | Tag: Very High***
- Q30.** Define 'Equilibrium' in biomechanics. Distinguish clearly between Static Equilibrium and Dynamic Equilibrium, providing one sports example for each. *Unit 8: Biomechanics and Sports | Tag: Very High***
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SECTION D: Case-Study Based Questions (12 Marks: 3 x 4)

All 3 questions are compulsory. Internal choice is provided in one sub-part of each.

- Q31.** The Physical Education Teacher of XYZ School is preparing a Knock-out fixture for an upcoming District Volleyball Tournament. A total of 13 teams have officially registered for the

competition. To ensure a fair draw, the teacher calculates the byes and matches on the whiteboard before drafting the bracket. **(i)** Calculate the total number of matches that will be played in this tournament. **(ii)** Calculate the total number of byes to be awarded in the first round of this fixture. **(iii)** How many teams will be placed in the lower half of the draw? **(iv)** Which specific mathematical formula is used to calculate the number of byes in the upper half? **OR** How many total rounds will be played in this tournament? *Unit 1: Management of Sporting Events | Tag: Very High***

Q32. Raj is a 14-year-old boy diagnosed with Autism Spectrum Disorder (ASD). He finds it difficult to interact with mainstream peers. His school sports coach introduces him to a specialized international movement that organizes sports training and athletic competitions exclusively for individuals with intellectual disabilities, aiming to build their self-confidence and social skills. **(i)** Which specific international organization/movement is the coach introducing Raj to? **(ii)** What does the abbreviation ASD stand for? **(iii)** Mention the minimum age required for an athlete to officially compete in this organization's games. **(iv)** Suggest any one inclusive strategy the coach can use to make the sports environment more accessible and comfortable for Raj. **OR** Which distinct organization governs and organizes games exclusively for hearing-impaired (deaf) athletes? *Unit 4: Physical Education & Sports for CWSN | Tag: Very High***

Q33. During a track and field training session, a Javelin thrower is analyzing his technique. The coach advises him: "To achieve the maximum distance, you need to release the javelin at a specific mathematical angle and apply explosive force. Remember, the harder you push your foot against the ground during the block, the more forward thrust your body generates for the throw." **(i)** The javelin flying freely through the air under the influence of gravity is scientifically termed a ___. **(ii)** What is the ideal mathematical angle of release to achieve the maximum horizontal distance? **(iii)** "The harder you push against the ground, the more forward thrust you generate" — this statement applies which of Newton's Laws of Motion? **(iv)** Name any one aerodynamic factor (other than gravity) that affects the javelin's flight trajectory. **OR** Newton's Second Law of Motion is also known as the Law of ___. *Unit 8: Biomechanics and Sports | Tag: Very High***

SECTION E: Long Answer Type (15 Marks: Attempt any 3 out of 4 x 5)

Word limit: 200-300 words.

Q34. What is a League Tournament? Draw a detailed and neat fixture of 7 teams based on a single league tournament using the Cyclic Method. Show all rounds clearly. *Unit 1: Management of Sporting Events | Tag: Very High***

Q35. Define Hypertension (High Blood Pressure). Name any four Yoga Asanas prescribed for its prevention and management. Thoroughly explain the step-by-step procedure and any two benefits of *Tadasana*. *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | Tag: Very High***

Q36. What do you understand by a 'Balanced Diet'? Elaborate comprehensively on the physiological functions and importance of any four Micro-Nutrients (specific Vitamins and Minerals) in maintaining an athlete's health and peak performance. *Unit 5: Sports & Nutrition* | Tag: High**

Q37. Soft tissue injuries are very common in high-intensity and contact sports. Define 'Soft Tissue Injuries'. Classify and thoroughly explain any four types of soft tissue injuries (such as Contusion, Abrasion, Laceration, Incision, Sprain, or Strain) detailing exactly how they occur on the field. *Unit 7: Physiology & Injuries in Sport* | Tag: Very High**

Answer Key

Q1. (B) League Tournament **Q2.** (B) $N - 1$ **Q3.** (C) Scoliosis **Q4.** (B) Amenorrhea **Q5.** (C) Asthma **Q6.** (B) Special Olympics **Q7.** (A) Sensory Processing Disorder **Q8.** (C) Vitamin B-Complex **Q9.** (C) Iron **Q10.** (C) Modified Push-ups **Q11.** (B) Lower body strength **Q12.** (B) Cardiac Output **Q13.** (B) Strain **Q14.** (A) Class I Lever **Q15.** (C) Rolling friction **Q16.** (C) (A) is true, but (R) is false. (Reason: A consolation tournament is a variation of a *Knock-out* format, not a League format). **Q17.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q18.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

Q19. 1. To identify and expose hidden athletic talent at higher competitive levels. 2. To build a sense of unity, brotherhood, and integration among students of different institutions. **Q20.** 1. Walking or running on toes and heels alternately. 2. Skipping a rope or trying to pick up small marbles from the floor using only the toes. **Q21.** 1. Individuals suffering from high blood pressure, cardiac ailments, or severe headaches should avoid it. 2. People with hernia, recent abdominal surgeries, or severe spinal injuries must not perform it. **Q22. Classification** is used in the Paralympics to group athletes based on the medical degree of their physical/visual impairment. **Divisoning** is uniquely used in the Special Olympics to group athletes based on age, gender, and their *current sports skill/ability level* to ensure fair competition. **Q23. Administration:** The athlete sits barefoot on the floor with legs fully extended and feet flat against a 'Sit & Reach' box. They slowly reach forward with both hands as far as possible along the measuring scale and hold the position for 2 seconds. The maximum reach distance is recorded in centimeters. **Q24. Vital Capacity:** The maximum volume of air a person can exhale after taking the deepest possible breath. **Residual Volume:** The volume of air that always remains in the lungs even after a maximal, forceful exhalation.

Q25. First Aid Committee: Responsible for providing immediate medical assistance to athletes who get injured during the event. They must keep medical kits ready and arrange ambulances or doctors for severe emergencies. **Finance Committee:** Responsible for all monetary transactions. They prepare the event budget, manage expenditures (equipment, boarding, prizes), and handle sponsorships and income records. **Q26.** 1. **Reduces Stress & Depression:** Physical activity releases endorphins, improving mood and reducing anxiety. 2.

Builds Self-Esteem: Mastering sports skills gives women confidence in their physical and mental capabilities. 3. **Fosters Independence & Leadership:** Sports teach goal-setting, resilience, and teamwork, empowering women to take on leadership roles in society. **Q27. Food Myths** are unfounded, unscientific beliefs about certain foods or diets. 1. **Myth: "Carbohydrates make you fat."** *Fact:* Excess caloric intake makes you fat. Complex carbs are essential for sustained energy. 2. **Myth: "Skipping meals reduces weight quickly."** *Fact:* Skipping meals lowers the Basal Metabolic Rate (BMR), leading to muscle loss and extreme fatigue rather than healthy fat loss. **Q28. Purpose:** To measure the coordination and speed of movement of the upper limbs. **Equipment:** A height-adjustable table, two yellow discs (20cm diameter), a center rectangle, and a stopwatch. **Procedure:** The child places their non-preferred hand on the center rectangle. With the preferred hand, they rapidly tap the two yellow discs alternately. The time taken to complete 25 full cycles (50 taps) is recorded. **Q29. 1. Muscular Hypertrophy:** The physical cross-sectional size and mass of the muscle fibers increase. 2. **Increased Mitochondria:** The size and number of mitochondria (the powerhouse of cells) multiply, massively boosting energy (ATP) production. 3. **Increase in Myoglobin:** Myoglobin content increases, allowing muscles to store more oxygen, delaying fatigue during long events. **Q30. Equilibrium** is a state of balance where all opposing forces acting on the body cancel each other out. **Static Equilibrium:** Balancing the body while completely at rest (e.g., a gymnast holding a steady handstand). **Dynamic Equilibrium:** Maintaining balance while the body is in uniform motion (e.g., a cyclist riding around a velodrome curve without falling).

Q31. (i) Total Matches = $N - 1 = 13 - 1 = 12$ matches. (ii) Total Byes = Next power of 2 minus $N = 16 - 13 = 3$ byes. (iii) Teams in lower half = $(N - 1)/2 = 12/2 = 6$ teams. (iv) Formula for byes in upper half: $(N_b - 1)/2$. **OR** Rounds = $2^4 = 16$, so 4 rounds will be played. **Q32.** (i) Special Olympics. (ii) Autism Spectrum Disorder. (iii) 8 years old. (iv) Use simple, clear, and very short instructions, and maintain a highly structured routine without sudden loud noises. **OR** ICSD (International Committee of Sports for the Deaf) / Deaflympics. **Q33.** (i) Projectile. (ii) 45° . (iii) Newton's Third Law of Motion (Law of Action and Reaction). (iv) Air resistance (friction) or Wind velocity. **OR** Law of Acceleration (or Momentum).

Q34. League Tournament: A format where every participating team plays against every other team, regardless of winning or losing previous matches. It determines the true best team. **Calculations for 7 Teams (N=7):** Matches = $7(6)/2 = 21$ matches. Rounds = $N = 7$ rounds (since N is odd, a 'Bye' is used). **Cyclic Fixture Draw:** In the Cyclic method, when N is odd, 'Bye' is fixed at the top right position. The teams rotate clockwise around the Bye in each round. * Round 1: 7-Bye, 6-1, 5-2, 4-3 * Round 2: 6-Bye, 5-7, 4-1, 3-2 * Round 3: 5-Bye, 4-6, 3-7, 2-1 * Round 4: 4-Bye, 3-5, 2-6, 1-7 * Round 5: 3-Bye, 2-4, 1-5, 7-6 * Round 6: 2-Bye, 1-3, 7-4, 6-5 * Round 7: 1-Bye, 7-2, 6-3, 5-4 (*Evaluator note: Full marks for correctly showing the clockwise rotation across all 7 rounds*). **Q35. Hypertension** is a condition where the force of the blood against the artery walls is too high (above 140/90 mmHg). **Asanas:** Tadasana, Katichakrasana, Uttanpadasana, Ardha Halasana, Shavasana. **Procedure of Tadasana (Palm Tree Pose):** Stand straight with feet together. Interlock the fingers and turn the palms upward over the head. Inhale deeply and stretch the arms straight up while simultaneously raising the heels,

balancing the entire body weight on the toes. Stretch the whole body upwards. Hold the pose for 10-15 seconds while breathing normally. Exhale and slowly bring the heels and arms down. **Benefits:** 1. Highly effective in reducing hypertension and calming the mind. 2. Cures postural deformities like flat foot and helps in increasing height in children. **Q36. A Balanced Diet** is one that contains all the essential macro-nutrients (carbs, proteins, fats) and micro-nutrients (vitamins, minerals) in the correct ratio required to maintain homeostasis and athletic efficiency. **Micro-Nutrients (Needed in small quantities):** 1. **Vitamin C (Water-soluble):** Essential for healing wounds, maintaining healthy connective tissues/ligaments, and boosting immunity against infections. 2. **Vitamin D (Fat-soluble):** Critical for the absorption of calcium in the intestines, directly ensuring strong bone density and preventing stress fractures in athletes. 3. **Calcium (Macro-mineral):** The fundamental mineral required for bone and teeth formation. It also plays a vital physiological role in muscle contraction mechanisms. 4. **Iron (Micro-mineral):** Crucial for the formation of hemoglobin in red blood cells. Adequate iron ensures maximum oxygen transport to working muscles, directly determining aerobic endurance. **Q37. Soft Tissue Injuries** involve damage to the skin, muscles, tendons, or ligaments, usually resulting from direct trauma or overexertion. 1. **Contusion (Bruise):** Caused by a direct blunt hit (e.g., collision with a player or equipment). Blood vessels rupture internally beneath the skin, causing a blue/black discoloration and swelling without breaking the skin layer. 2. **Abrasion:** A surface-level skin injury caused by friction or scraping against a rough surface (e.g., a football player sliding and scraping their knee on artificial turf). 3. **Sprain:** A severe injury to a **ligament** (bone-to-bone connective tissue). It occurs when a joint is violently twisted or wrenched beyond its normal range of motion, commonly seen in the ankle or wrist. 4. **Strain:** A severe injury to a **muscle** or **tendon** (muscle-to-bone). Also known as a "muscle pull," it occurs due to overstretching or violent contraction of muscles, very commonly seen as hamstring strains in sprinters.

CBSE Class 12 (2027) — Half-Yearly Predicted Paper (80% syllabus). AI-generated via PYQ/Blueprint analysis, not actual exam questions.